

HW 1 : 1D Gaussian Filtering

과목 : 기초 GPU 프로그래밍

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GPU : Geforce RTX 4070 Ti

CUDA Compute Capability : 8.9

측정 방법

GPU kernel 함수 `apply_1D_Gaussian_device`와 Host 함수 `apply_1D_Gaussian_host`가 1d Gaussian Filtering 수행 시간에 얼마나 차이가 나는지 확인하기 위해 아래와 같은 변화를 주었다.

방법 1 : host, kernel 함수에 대해서 입력 영상 크기, 필터 너비, thread block 크기를 변화시켜 가면서 host(cpu)의 필터링 소요 시간, device(gpu)의 필터링 소요 시간, 평균 소요 시간을 측정했다.

방법 2 : 방법 1에서의 kernel 함수가 grid-stride loop 기능을 제공하게 함수를 수정하고 GRID_REDUCTION_FACTOR 값을 변경하면서 소요 시간에 변화가 있는지 측정했다.

이때, 방법 2는 필터 너비를 11로 고정하여 GRID REDUCTION FACTOR 값 변화에 따른 소요 시간 변화를 측정하였다.

입력 영상 크기(NUMBER_OF_ELEMENTS) : $2^{20}, 2^{22}, 2^{24}, 2^{26}$

필터 너비(kernel_width) : 1, 3, 5, 7, 9, 11

Thread block 크기(THREADS_PER_BLOCK) : 64, 128, 256, 512, 1024

GRID REDUCTION FACTOR(방법 2, Device 한정) : 1, 2, 4, 8

- 위의 모든 실험 조건을 조합하여 측정 결과를 얻었다.

float 자료형 연산 후 int로 형 변환 과정에서 cpu, gpu 아키텍처 차이로 미세한 오차가 발생할 수 있기에

$$\text{abs}(h_image_out_host[i] - h_image_out_device[i]) > 1$$

로 허용 오차 범위를 설정해 검증했다.

측정 결과 및 분석

방법 1 (HOST)	입력 영상 크기 NUMBER OF ELEMENTS	필터 너비 kernel_width	Thread block 크기 THREADS_PER_BLOCK	GRID REDUCTION FACTOR	Time (ms)
HOST	2^20	1	64	1	2.191
	2^20	3	64	1	3.104
	2^20	5	64	1	3.975
	2^20	7	64	1	5.817
	2^20	9	64	1	6.907
	2^20	11	64	1	9.667
HOST	2^20	1	128	1	2.551
	2^20	3	128	1	3.210
	2^20	5	128	1	4.155
	2^20	7	128	1	6.001
	2^20	9	128	1	6.647
	2^20	11	128	1	8.437
HOST	2^20	1	256	1	2.483
	2^20	3	256	1	3.170
	2^20	5	256	1	3.963
	2^20	7	256	1	5.753
	2^20	9	256	1	6.748
	2^20	11	256	1	8.733
HOST	2^20	1	512	1	2.246
	2^20	3	512	1	3.309
	2^20	5	512	1	4.198
	2^20	7	512	1	5.899
	2^20	9	512	1	7.219
	2^20	11	512	1	8.316
HOST	2^20	1	1024	1	2.438
	2^20	3	1024	1	3.062
	2^20	5	1024	1	4.021
	2^20	7	1024	1	5.791
	2^20	9	1024	1	7.320
	2^20	11	1024	1	8.503
HOST	2^22	1	64	1	8.114
	2^22	3	64	1	15.585
	2^22	5	64	1	22.283
	2^22	7	64	1	23.531
	2^22	9	64	1	29.279
	2^22	11	64	1	37.388
HOST	2^22	1	128	1	7.75
	2^22	3	128	1	12.659
	2^22	5	128	1	16.550
	2^22	7	128	1	24.140
	2^22	9	128	1	27.375

	2^22	11	128	1	34.129
HOST	2^22	1	256	1	8.571
	2^22	3	256	1	13.076
	2^22	5	256	1	16.694
	2^22	7	256	1	25.596
	2^22	9	256	1	27.260
	2^22	11	256	1	35.464
HOST	2^22	1	512	1	7.912
	2^22	3	512	1	13.370
	2^22	5	512	1	16.440
	2^22	7	512	1	22.904
	2^22	9	512	1	26.753
	2^22	11	512	1	34.228
HOST	2^22	1	1024	1	7.695
	2^22	3	1024	1	12.719
	2^22	5	1024	1	16.053
	2^22	7	1024	1	23.839
	2^22	9	1024	1	28.564
	2^22	11	1024	1	36.38
HOST	2^24	1	64	1	35.665
	2^24	3	64	1	51.468
	2^24	5	64	1	90.905
	2^24	7	64	1	116.135
	2^24	9	64	1	113.254
	2^24	11	64	1	147.88
HOST	2^24	1	128	1	31.448
	2^24	3	128	1	51.431
	2^24	5	128	1	63.778
	2^24	7	128	1	92.671
	2^24	9	128	1	107.073
	2^24	11	128	1	135.408
HOST	2^24	1	256	1	33.97
	2^24	3	256	1	52.574
	2^24	5	256	1	65.190
	2^24	7	256	1	96.516
	2^24	9	256	1	155.605
	2^24	11	256	1	160.648
HOST	2^24	1	512	1	34.166
	2^24	3	512	1	52.051
	2^24	5	512	1	100.189
	2^24	7	512	1	132.081
	2^24	9	512	1	126.889
	2^24	11	512	1	139.4
HOST	2^24	1	1024	1	31.071
	2^24	3	1024	1	49.554
	2^24	5	1024	1	64.919

	2 ²⁴	7	1024	1	92.583
	2 ²⁴	9	1024	1	107.657
	2 ²⁴	11	1024	1	135.237
HOST	2 ²⁶	1	64	1	188.881
	2 ²⁶	3	64	1	216.850
	2 ²⁶	5	64	1	266.918
	2 ²⁶	7	64	1	375.685
	2 ²⁶	9	64	1	437.388
	2 ²⁶	11	64	1	553.325
HOST	2 ²⁶	1	128	1	127.115
	2 ²⁶	3	128	1	277.314
	2 ²⁶	5	128	1	279.384
	2 ²⁶	7	128	1	393.409
	2 ²⁶	9	128	1	433.331
	2 ²⁶	11	128	1	558.936
HOST	2 ²⁶	1	256	1	165.191
	2 ²⁶	3	256	1	214.584
	2 ²⁶	5	256	1	270.794
	2 ²⁶	7	256	1	379.934
	2 ²⁶	9	256	1	441.985
	2 ²⁶	11	256	1	552.954
HOST	2 ²⁶	1	512	1	177.861
	2 ²⁶	3	512	1	214.428
	2 ²⁶	5	512	1	267.829
	2 ²⁶	7	512	1	371.966
	2 ²⁶	9	512	1	433.014
	2 ²⁶	11	512	1	550.378
HOST	2 ²⁶	1	1024	1	146.144
	2 ²⁶	3	1024	1	227.921
	2 ²⁶	5	1024	1	271.930
	2 ²⁶	7	1024	1	380.319
	2 ²⁶	9	1024	1	430.818
	2 ²⁶	11	1024	1	562.557

<표 1 : [방법 1] 1D Gaussain Filtering Host 수행 시간>

방법 1 (DEVICE)	입력 영상 크기 NUMBER OF ELEMENTS	필터 너비 kernel_width	Thread block 크기 THREADS_PER_BLOCK	GRID REDUCTION FACTOR	Time (ms)	Avg Time (ms)
DEVICE	2^20	1	64	1	0.055	0.012
	2^20	3	64	1	0.078	0.012
	2^20	5	64	1	0.082	0.012
	2^20	7	64	1	0.083	0.012
	2^20	9	64	1	0.090	0.015
	2^20	11	64	1	0.208	0.020
DEVICE	2^20	1	128	1	0.038	0.008
	2^20	3	128	1	0.069	0.009
	2^20	5	128	1	0.247	0.013
	2^20	7	128	1	0.265	0.013
	2^20	9	128	1	0.090	0.016
	2^20	11	128	1	0.191	0.017
DEVICE	2^20	1	256	1	0.202	0.018
	2^20	3	256	1	0.070	0.01
	2^20	5	256	1	0.086	0.017
	2^20	7	256	1	0.123	0.052
	2^20	9	256	1	0.093	0.021
	2^20	11	256	1	0.088	0.017
DEVICE	2^20	1	512	1	0.024	0.009
	2^20	3	512	1	0.200	0.035
	2^20	5	512	1	0.089	0.021
	2^20	7	512	1	0.180	0.014
	2^20	9	512	1	0.080	0.016
	2^20	11	512	1	0.093	0.018
DEVICE	2^20	1	1024	1	0.028	0.012
	2^20	3	1024	1	0.075	0.014
	2^20	5	1024	1	0.095	0.016
	2^20	7	1024	1	0.094	0.017
	2^20	9	1024	1	0.102	0.042
	2^20	11	1024	1	0.113	0.021
DEVICE	2^22	1	64	1	0.140	0.036
	2^22	3	64	1	0.206	0.035
	2^22	5	64	1	0.206	0.036
	2^22	7	64	1	0.219	0.037
	2^22	9	64	1	0.257	0.05
	2^22	11	64	1	0.226	0.051
DEVICE	2^22	1	128	1	0.132	0.025
	2^22	3	128	1	0.132	0.024
	2^22	5	128	1	0.193	0.036
	2^22	7	128	1	0.216	0.037
	2^22	9	128	1	0.225	0.05
	2^22	11	128	1	0.228	0.05
DEVICE	2^22	1	256	1	0.139	0.024

	2^22	3	256	1	0.228	0.025
	2^22	5	256	1	0.203	0.036
	2^22	7	256	1	0.224	0.037
	2^22	9	256	1	0.215	0.052
	2^22	11	256	1	0.228	0.089
DEVICE	2^22	1	512	1	0.143	0.024
	2^22	3	512	1	0.138	0.025
	2^22	5	512	1	0.202	0.064
	2^22	7	512	1	0.400	0.037
	2^22	9	512	1	0.230	0.052
	2^22	11	512	1	0.170	0.051
DEVICE	2^22	1	1024	1	0.164	0.095
	2^22	3	1024	1	0.169	0.038
	2^22	5	1024	1	0.247	0.079
	2^22	7	1024	1	0.247	0.089
	2^22	9	1024	1	0.268	0.077
	2^22	11	1024	1	0.264	0.136
DEVICE	2^24	1	64	1	0.628	0.438
	2^24	3	64	1	0.661	0.432
	2^24	5	64	1	1.272	0.326
	2^24	7	64	1	0.691	0.484
	2^24	9	64	1	1.075	0.321
	2^24	11	64	1	0.665	0.432
DEVICE	2^24	1	128	1	0.56	0.451
	2^24	3	128	1	0.655	0.433
	2^24	5	128	1	0.741	0.426
	2^24	7	128	1	0.876	0.312
	2^24	9	128	1	0.692	0.373
	2^24	11	128	1	0.731	0.428
DEVICE	2^24	1	256	1	0.985	0.316
	2^24	3	256	1	1.469	0.342
	2^24	5	256	1	0.696	0.409
	2^24	7	256	1	0.709	0.377
	2^24	9	256	1	0.692	0.431
	2^24	11	256	1	0.697	0.436
DEVICE	2^24	1	512	1	0.611	0.42
	2^24	3	512	1	0.649	0.447
	2^24	5	512	1	1.240	0.396
	2^24	7	512	1	0.683	0.314
	2^24	9	512	1	1.179	0.307
	2^24	11	512	1	1.395	0.308
DEVICE	2^24	1	1024	1	1.109	0.357
	2^24	3	1024	1	0.688	0.357
	2^24	5	1024	1	0.748	0.321
	2^24	7	1024	1	0.759	0.38
	2^24	9	1024	1	0.812	0.435

	2^24	11	1024	1	0.773	0.447
DEVICE	2^26	1	64	1	2.481	1.468
	2^26	3	64	1	2.551	1.426
	2^26	5	64	1	3.065	1.463
	2^26	7	64	1	2.550	1.539
	2^26	9	64	1	3.272	1.515
	2^26	11	64	1	3.244	1.421
DEVICE	2^26	1	128	1	4.174	1.532
	2^26	3	128	1	3.226	1.466
	2^26	5	128	1	3.689	1.419
	2^26	7	128	1	3.717	1.506
	2^26	9	128	1	3.689	1.487
	2^26	11	128	1	3.429	1.447
DEVICE	2^26	1	256	1	2.846	1.441
	2^26	3	256	1	2.609	1.394
	2^26	5	256	1	2.879	1.399
	2^26	7	256	1	3.148	1.452
	2^26	9	256	1	2.596	1.413
	2^26	11	256	1	2.583	1.431
DEVICE	2^26	1	512	1	2.905	1.518
	2^26	3	512	1	3.008	1.433
	2^26	5	512	1	2.548	1.45
	2^26	7	512	1	2.614	1.437
	2^26	9	512	1	2.548	1.447
	2^26	11	512	1	3.445	1.476
DEVICE	2^26	1	1024	1	2.747	1.464
	2^26	3	1024	1	3.127	1.473
	2^26	5	1024	1	3.973	1.516
	2^26	7	1024	1	2.587	1.565
	2^26	9	1024	1	3.539	1.578
	2^26	11	1024	1	3.77	1.571

<표 2 : [방법 1] 1D Gaussain Filtering Device 수행 시간>

방법 2 (DEVICE)	입력 영상 크기 NUMBER OF ELEMENTS	필터 너비 kernel_width	Thread block 크기 THREADS_PER_BLOCK	GRID REDUCTION FACTOR	Time (ms)	Avg Time (ms)
DEVICE	2^20	11	64	1	0.083	0.017
	2^20	11	64	2	0.149	0.016
	2^20	11	64	4	0.085	0.016
	2^20	11	64	8	0.153	0.042

DEVICE	2^20	11	128	1	0.09	0.049
	2^20	11	128	2	0.093	0.016
	2^20	11	128	4	0.092	0.017
	2^20	11	128	8	0.086	0.016

DEVICE	2^20	11	256	1	0.092	0.016
	2^20	11	256	2	0.155	0.068
	2^20	11	256	4	0.154	0.016
	2^20	11	256	8	0.092	0.017

DEVICE	2^20	11	512	1	0.089	0.017
	2^20	11	512	2	0.153	0.017
	2^20	11	512	4	0.152	0.016
	2^20	11	512	8	0.165	0.027

DEVICE	2^20	11	1024	1	0.098	0.02
	2^20	11	1024	2	0.518	0.027
	2^20	11	1024	4	0.197	0.019
	2^20	11	1024	8	0.079	0.019

DEVICE	2^22	11	64	1	0.289	0.105
	2^22	11	64	2	0.155	0.185
	2^22	11	64	4	0.219	0.049
	2^22	11	64	8	0.247	0.048

DEVICE	2^22	11	128	1	0.335	0.106
	2^22	11	128	2	0.221	0.08
	2^22	11	128	4	0.189	0.171
	2^22	11	128	8	0.224	0.053

DEVICE	2^22	11	256	1	0.213	0.076
	2^22	11	256	2	0.233	0.05
	2^22	11	256	4	0.232	0.049
	2^22	11	256	8	0.21	0.067

DEVICE	2^22	11	512	1	0.346	0.12
	2^22	11	512	2	0.157	0.049
	2^22	11	512	4	0.24	0.048
	2^22	11	512	8	0.216	0.076

DEVICE	2^22	11	1024	1	0.208	0.067
	2^22	11	1024	2	0.289	0.057
	2^22	11	1024	4	0.232	0.053
	2^22	11	1024	8	0.168	0.076

DEVICE	2^24	11	64	1	0.74	0.389
	2^24	11	64	2	0.693	0.406
	2^24	11	64	4	0.708	0.447
	2^24	11	64	8	0.725	0.405

DEVICE	2^24	11	128	1	1	0.328
	2^24	11	128	2	0.703	0.31
	2^24	11	128	4	1.267	0.325
	2^24	11	128	8	0.956	0.407

DEVICE	2^24	11	256	1	0.743	0.395
	2^24	11	256	2	1.359	0.355
	2^24	11	256	4	1.187	0.314
	2^24	11	256	8	0.708	0.441

DEVICE	2^24	11	512	1	0.668	0.422
	2^24	11	512	2	1.365	0.318
	2^24	11	512	4	0.697	0.337
	2^24	11	512	8	0.704	0.352

DEVICE	2^24	11	1024	1	0.785	0.449
	2^24	11	1024	2	0.754	0.462
	2^24	11	1024	4	0.955	0.402

	2^24	11	1024	8	0.947	0.403
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DEVICE	2^26	11	64	1	2.539	1.463
	2^26	11	64	2	3.014	1.444
	2^26	11	64	4	2.554	1.458
	2^26	11	64	8	2.565	1.409

DEVICE	2^26	11	128	1	3.279	1.376
	2^26	11	128	2	3.437	1.487
	2^26	11	128	4	3.437	1.512
	2^26	11	128	8	3.6	1.488

DEVICE	2^26	11	256	1	2.577	1.414
	2^26	11	256	2	2.597	1.37
	2^26	11	256	4	2.618	1.342
	2^26	11	256	8	2.598	1.403

DEVICE	2^26	11	512	1	2.958	1.479
	2^26	11	512	2	2.584	1.466
	2^26	11	512	4	2.552	1.468
	2^26	11	512	8	2.592	1.514

DEVICE	2^26	11	1024	1	2.732	1.7
	2^26	11	1024	2	3.123	1.492
	2^26	11	1024	4	3.324	1.409
	2^26	11	1024	8	3.235	1.365

<표 3 : [방법 2] 1D Gaussain Filtering Host 수행 시간>

[방법 1] 분석

Host 결과 분석

- 입력 영상 크기가 커질수록 처리해야할 데이터가 커지며 for loop의 반복 회수가 증가하기에 수행 시간이 정비례하며 증가했다.
- 필터 너비가 증가할수록 픽셀 당 수행해야하는 곱셈과 덧셈 횟수가 선형적으로 늘어나기에 수행 시간이 선형적으로 증가했다.
- Thread block 크기가 증가해도 수행 시간에는 큰 차이가 없었다. Host 함수는 CPU의 단일 코어에서 Sequential 하게 실행되기 때문에 이와 같은 결과가 얻어진 것으로 보인다.

Device 결과 분석

- 입력 영상 크기가 커질수록 처리해야할 데이터가 Host와 마찬가지로 커졌지만 GPU는 수많은 코어에 작업을 parallel하게 수행하기에 시간 증가 비율이 Host에 비해 작았다.
- 필터 너비가 커져도 평균 수행 시간의 증가폭은 Host 비해서 많이 작았다. 이도 마찬가지로 수많은 스레드가 덧셈과 곱셈을 동시에 parallel하게 수행해서 나온 결과로 분석된다.
- Thread block 크기 변화도 마찬가지로 평균 수행 시간에 큰 차이가 나타나지 않았다. GPU의 register와 shared memory 덕분에 병목이 발생하지 않고 높은 warp occupancy가 달성되었기 때문이라고 판단된다.

Host와 Device 비교 결과 분석

- CPU는 모든 픽셀을 직렬처리하지만 GPU는 수많은 thread가 픽셀을 하나씩 parallel하게 처리하기에 Host와 Device 간의 수행 시간 차이가 컸다. 또한, 입력 영상 크기나 필터 너비가 증가해 연산량이 증가할수록 Host 수행 시간은 최대 562.557ms까지 늘어났지만 GPU는 최대 1.571ms에 그친 것을 보아 GPU는 Parallel 처리 효율이 좋아 안정적인 수행 시간을 보인 것 같다.

[방법 2] 분석 (필터 너비 11로 고정)

- 입력 영상 크기가 커질수록 각 thread가 처리하는 loop 횟수가 증가하기에 수행 시간이 영상 크기에 비례해서 증가했다.
- Thread Block 크기가 증가해도 중간중간 오차 값을 제외하면 평균 수행 시간 변화가 거의 없었다. GPU의 자원 할당 능력 덕분에 블록 크기가 커져도 수행 시간에서 큰 차이가 없던 것 같다.
- GRID_REDUCTION_FACTOR 값이 1, 2, 4, 8로 변화하면 전체 thread block 수가 1, 1/2, 1/4, 1/8로 줄어든다. 하지만 측정 결과에서는 오차값을 제외하고 평균 수행 시간은 거의 일정하게 유지되었다. 이것은 Grid-stride loop 구조가 thread가 한 픽셀을 처리하고 제거되는 대신 다음 인덱스로 점프하여 sequential 하게 연산을 수행하기 때문으로 보인다. 이 덕분에 thread를 생성하고 제거하는데에 소요되는 overhead가 줄어들고 cache 활용도가 극대화 되었기에 이런 결과가 나온 것으로 보인다.

아래는 프로그램 실행 결과들이다.

```

[Method 2] Started
--- GRID_REDUCTION_FACTOR = 1 ---
*** CUDA kernel launch with 16384(16384) blocks of 64 threads
*** Time to filter an image of 1048576 pixels on the device = 0.083(ms)
*** Average time to filter an image of 1048576 pixels on the device = 0.017(ms)
*** Verifying the result from the device...
*** Test PASSED
[1D Gaussian filtering of image of 1048576 pixels(KERNEL WIDTH = 3)]
*** Time to filter an image of 1048576 pixels on the host = 3.104(ms)
*** CUDA kernel launch with 16384(16384) blocks of 64 threads
*** Time to filter an image of 1048576 pixels on the device = 0.078(ms)
*** Average time to filter an image of 1048576 pixels on the device = 0.012(ms)
*** Verifying the result from the device...
*** Test PASSED
[1D Gaussian filtering of image of 1048576 pixels(KERNEL WIDTH = 5)]
*** Time to filter an image of 1048576 pixels on the host = 3.975(ms)
*** CUDA kernel launch with 16384(16384) blocks of 64 threads
*** Time to filter an image of 1048576 pixels on the device = 0.082(ms)
*** Average time to filter an image of 1048576 pixels on the device = 0.012(ms)
*** Verifying the result from the device...
*** Test PASSED
[1D Gaussian filtering of image of 1048576 pixels(KERNEL WIDTH = 7)]
*** Time to filter an image of 1048576 pixels on the host = 5.817(ms)
*** CUDA kernel launch with 16384(16384) blocks of 64 threads
*** Time to filter an image of 1048576 pixels on the device = 0.083(ms)
*** Average time to filter an image of 1048576 pixels on the device = 0.012(ms)
*** Verifying the result from the device...
*** Test PASSED
[1D Gaussian filtering of image of 1048576 pixels(KERNEL WIDTH = 9)]
*** Time to filter an image of 1048576 pixels on the host = 6.907(ms)
*** CUDA kernel launch with 16384(16384) blocks of 64 threads
*** Time to filter an image of 1048576 pixels on the device = 0.090(ms)
*** Average time to filter an image of 1048576 pixels on the device = 0.015(ms)
*** Verifying the result from the device...
*** Test PASSED
[1D Gaussian filtering of image of 1048576 pixels(KERNEL WIDTH = 11)]
*** Time to filter an image of 1048576 pixels on the host = 9.607(ms)
*** CUDA kernel launch with 16384(16384) blocks of 64 threads
*** Time to filter an image of 1048576 pixels on the device = 0.208(ms)
*** Average time to filter an image of 1048576 pixels on the device = 0.020(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 2 ---
*** CUDA kernel launch with 8192(16384) blocks of 64 threads
*** Time to filter an image of 1048576 pixels on the device = 0.149(ms)
*** Average time to filter an image of 1048576 pixels on the device = 0.016(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 4 ---
*** CUDA kernel launch with 4096(16384) blocks of 64 threads
*** Time to filter an image of 1048576 pixels on the device = 0.085(ms)
*** Average time to filter an image of 1048576 pixels on the device = 0.016(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 8 ---
*** CUDA kernel launch with 2048(16384) blocks of 64 threads
*** Time to filter an image of 1048576 pixels on the device = 0.153(ms)
*** Average time to filter an image of 1048576 pixels on the device = 0.042(ms)
*** Verifying the result from the device...
*** Test PASSED
*** Done

```

```
[Method 2] Started -----  
-- GRID_REDUCTION_FACTOR = 1 --  
  
*** Gaussian filtering of image of 1048576 pixels(CHANNEL WIDTH = 1)  
*** Time to filter an image of 1048576 pixels on the host = 2.51(ms)  
*** CUDA kernel Launch with 8192(8192) blocks of 128 threads  
*** Time to filter an image of 1048576 pixels on the device = 0.618(ms)  
*** Average time to filter an image of 1048576 pixels on the device = 0.888(ms)  
*** Verifying the result from the device...  
*** Test PASSED  
  
[ID Gaussian filtering of image of 1048576 pixels(CHANNEL WIDTH = 3)]  
*** Time to filter an image of 1048576 pixels on the host = 3.216(ms)  
*** CUDA kernel Launch with 8192(8192) blocks of 128 threads  
*** Time to filter an image of 1048576 pixels on the device = 0.969(ms)  
*** Average time to filter an image of 1048576 pixels on the device = 0.869(ms)  
*** Verifying the result from the device...  
*** Test PASSED  
  
[ID Gaussian filtering of image of 1048576 pixels(CHANNEL WIDTH = 5)]  
*** Time to filter an image of 1048576 pixels on the host = 4.155(ms)  
*** CUDA kernel Launch with 8192(8192) blocks of 128 threads  
*** Time to filter an image of 1048576 pixels on the device = 0.207(ms)  
*** Average time to filter an image of 1048576 pixels on the device = 0.813(ms)  
*** Verifying the result from the device...  
*** Test PASSED  
  
[ID Gaussian filtering of image of 1048576 pixels(CHANNEL WIDTH = 7)]  
*** Time to filter an image of 1048576 pixels on the host = 6.061(ms)  
*** CUDA kernel Launch with 8192(8192) blocks of 128 threads  
*** Time to filter an image of 1048576 pixels on the device = 0.265(ms)  
*** Average time to filter an image of 1048576 pixels on the device = 0.813(ms)  
*** Verifying the result from the device...  
*** Test PASSED  
  
[ID Gaussian filtering of image of 1048576 pixels(CHANNEL WIDTH = 9)]  
*** Time to filter an image of 1048576 pixels on the host = 6.647(ms)  
*** CUDA kernel Launch with 8192(8192) blocks of 128 threads  
*** Time to filter an image of 1048576 pixels on the device = 0.696(ms)  
*** Average time to filter an image of 1048576 pixels on the device = 0.816(ms)  
*** Verifying the result from the device...  
*** Test PASSED  
  
[ID Gaussian filtering of image of 1048576 pixels(CHANNEL WIDTH = 11)]  
*** Time to filter an image of 1048576 pixels on the host = 8.437(ms)  
*** CUDA kernel Launch with 8192(8192) blocks of 128 threads  
*** Time to filter an image of 1048576 pixels on the device = 0.191(ms)  
*** Average time to filter an image of 1048576 pixels on the device = 0.817(ms)  
*** Verifying the result from the device...  
*** Test PASSED
```



```

[Method 2] Started
*** Time to filter an image of 16777216 pixels on the host = 31.408(ms)
*** CUDA kernel launch with 131072(131072) blocks of 128 threads
*** Time to filter an image of 16777216 pixels on the device = 0.566(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.451(ms)
*** Verifying the result from the device...
*** Test PASSED
[1D Gaussian filtering of image of 16777216 pixels(GERNEL WIDTH = 3)]
*** Time to filter an image of 16777216 pixels on the host = 51.431(ms)
*** CUDA kernel launch with 131072(131072) blocks of 128 threads
*** Time to filter an image of 16777216 pixels on the device = 0.655(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.433(ms)
*** Verifying the result from the device...
*** Test PASSED
[1D Gaussian filtering of image of 16777216 pixels(GERNEL WIDTH = 5)]
*** Time to filter an image of 16777216 pixels on the host = 61.778(ms)
*** CUDA kernel launch with 131072(131072) blocks of 128 threads
*** Time to filter an image of 16777216 pixels on the device = 0.701(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.426(ms)
*** Verifying the result from the device...
*** Test PASSED
[1D Gaussian filtering of image of 16777216 pixels(GERNEL WIDTH = 7)]
*** Time to filter an image of 16777216 pixels on the host = 92.071(ms)
*** CUDA kernel launch with 131072(131072) blocks of 128 threads
*** Time to filter an image of 16777216 pixels on the device = 0.876(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.312(ms)
*** Verifying the result from the device...
*** Test PASSED
[1D Gaussian filtering of image of 16777216 pixels(GERNEL WIDTH = 9)]
*** Time to filter an image of 16777216 pixels on the host = 107.073(ms)
*** CUDA kernel launch with 131072(131072) blocks of 128 threads
*** Time to filter an image of 16777216 pixels on the device = 0.692(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.373(ms)
*** Verifying the result from the device...
*** Test PASSED
[1D Gaussian filtering of image of 16777216 pixels(GERNEL WIDTH = 11)]
*** Time to filter an image of 16777216 pixels on the host = 115.408(ms)
*** CUDA kernel launch with 131072(131072) blocks of 128 threads
*** Time to filter an image of 16777216 pixels on the device = 0.731(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.428(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 1 ---
*** CUDA kernel launch with 131072(131072) blocks of 128 threads
*** Time to filter an image of 16777216 pixels on the device = 1.000(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.328(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 2 ---
*** CUDA kernel launch with 65536(131072) blocks of 128 threads
*** Time to filter an image of 16777216 pixels on the device = 0.703(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.310(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 4 ---
*** CUDA kernel launch with 32768(131072) blocks of 128 threads
*** Time to filter an image of 16777216 pixels on the device = 1.267(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.325(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 8 ---
*** CUDA kernel launch with 16384(131072) blocks of 128 threads
*** Time to filter an image of 16777216 pixels on the device = 0.956(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.407(ms)
*** Verifying the result from the device...
*** Test PASSED
*** Done

```

```

[Method 2] Started
--- GRID_REDUCTION_FACTOR = 1 ---
*** CUDA kernel launch with 65536(65536) blocks of 256 threads
*** Time to filter an image of 16777216 pixels on the device = 0.985(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.316(ms)
*** Verifying the result from the device...
*** Test PASSED
[GD Gaussian filtering of image of 16777216 pixels(CHANNEL WIDTH = 3)]
*** Time to filter an image of 16777216 pixels on the host = 52.574(ms)
*** CUDA kernel launch with 65536(65536) blocks of 256 threads
*** Time to filter an image of 16777216 pixels on the device = 1.469(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.342(ms)
*** Verifying the result from the device...
*** Test PASSED
[GD Gaussian filtering of image of 16777216 pixels(CHANNEL WIDTH = 5)]
*** Time to filter an image of 16777216 pixels on the host = 65.190(ms)
*** CUDA kernel launch with 65536(65536) blocks of 256 threads
*** Time to filter an image of 16777216 pixels on the device = 0.696(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.409(ms)
*** Verifying the result from the device...
*** Test PASSED
[GD Gaussian filtering of image of 16777216 pixels(CHANNEL WIDTH = 7)]
*** Time to filter an image of 16777216 pixels on the host = 96.516(ms)
*** CUDA kernel launch with 65536(65536) blocks of 256 threads
*** Time to filter an image of 16777216 pixels on the device = 0.709(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.377(ms)
*** Verifying the result from the device...
*** Test PASSED
[GD Gaussian filtering of image of 16777216 pixels(CHANNEL WIDTH = 9)]
*** Time to filter an image of 16777216 pixels on the host = 155.605(ms)
*** CUDA kernel launch with 65536(65536) blocks of 256 threads
*** Time to filter an image of 16777216 pixels on the device = 0.692(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.431(ms)
*** Verifying the result from the device...
*** Test PASSED
[GD Gaussian filtering of image of 16777216 pixels(CHANNEL WIDTH = 11)]
*** Time to filter an image of 16777216 pixels on the host = 168.608(ms)
*** CUDA kernel launch with 65536(65536) blocks of 256 threads
*** Time to filter an image of 16777216 pixels on the device = 0.697(ms)
*** Average time to filter an image of 16777216 pixels on the device = 0.436(ms)
*** Verifying the result from the device...
*** Test PASSED

```

```
[Method 2] Started -----
```

```
[D Gaussian filtering of image of 16777216 pixels][KERNEL WIDTH = 1]]  
*** Time to filter an image of 16777216 pixels on the host = 30.166(ms)  
--- GRID_REDUCTION_FACTOR = 1 ---  
*** CUDA kernel launch with 32768(32768) blocks of 512 threads  
*** Time to filter an image of 16777216 pixels on the device = 0.611(ms)  
*** Average time to filter an image of 16777216 pixels on the device = 0.428(ms)  
*** Verifying the result from the device...  
*** Test PASSED  
[D Gaussian filtering of image of 16777216 pixels][KERNEL WIDTH = 3]]  
*** Time to filter an image of 16777216 pixels on the host = 52.051(ms)  
--- GRID_REDUCTION_FACTOR = 2 ---  
*** CUDA kernel launch with 32768(32768) blocks of 512 threads  
*** Time to filter an image of 16777216 pixels on the device = 0.609(ms)  
*** Average time to filter an image of 16777216 pixels on the device = 0.407(ms)  
*** Verifying the result from the device...  
*** Test PASSED  
[D Gaussian filtering of image of 16777216 pixels][KERNEL WIDTH = 5]]  
*** Time to filter an image of 16777216 pixels on the host = 100.189(ms)  
--- GRID_REDUCTION_FACTOR = 4 ---  
*** CUDA kernel launch with 32768(32768) blocks of 512 threads  
*** Time to filter an image of 16777216 pixels on the device = 1.208(ms)  
*** Average time to filter an image of 16777216 pixels on the device = 0.396(ms)  
*** Verifying the result from the device...  
*** Test PASSED  
[D Gaussian filtering of image of 16777216 pixels][KERNEL WIDTH = 7]]  
*** Time to filter an image of 16777216 pixels on the host = 132.681(ms)  
--- GRID_REDUCTION_FACTOR = 8 ---  
*** CUDA kernel launch with 32768(32768) blocks of 512 threads  
*** Time to filter an image of 16777216 pixels on the device = 0.683(ms)  
*** Average time to filter an image of 16777216 pixels on the device = 0.316(ms)  
*** Verifying the result from the device...  
*** Test PASSED  
[D Gaussian filtering of image of 16777216 pixels][KERNEL WIDTH = 9]]  
*** Time to filter an image of 16777216 pixels on the host = 126.889(ms)  
--- GRID_REDUCTION_FACTOR = 8 ---  
*** CUDA kernel launch with 32768(32768) blocks of 512 threads  
*** Time to filter an image of 16777216 pixels on the device = 1.179(ms)  
*** Average time to filter an image of 16777216 pixels on the device = 0.397(ms)  
*** Verifying the result from the device...  
*** Test PASSED  
[D Gaussian filtering of image of 16777216 pixels][KERNEL WIDTH = 11]]  
*** Time to filter an image of 16777216 pixels on the host = 139.400(ms)  
--- GRID_REDUCTION_FACTOR = 8 ---  
*** CUDA kernel launch with 32768(32768) blocks of 512 threads  
*** Time to filter an image of 16777216 pixels on the device = 1.396(ms)  
*** Average time to filter an image of 16777216 pixels on the device = 0.388(ms)  
*** Verifying the result from the device...  
*** Test PASSED  
Don't Done
```



```

[1D Gaussian filtering of image of 67188864 pixels(KERNEL WIDTH = 1)]
*** Time to filter an image of 67188864 pixels on the host = 177.861(ms)
*** CUDA kernel launch with 131072(131072) blocks of 512 threads
*** Time to filter an image of 67188864 pixels on the device = 2.905(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.518(ms)
*** Verifying the result from the device...
*** Test PASSED

[1D Gaussian filtering of image of 67188864 pixels(KERNEL WIDTH = 3)]
*** Time to filter an image of 67188864 pixels on the host = 214.428(ms)
*** CUDA kernel launch with 131072(131072) blocks of 512 threads
*** Time to filter an image of 67188864 pixels on the device = 3.088(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.433(ms)
*** Verifying the result from the device...
*** Test PASSED

[1D Gaussian filtering of image of 67188864 pixels(KERNEL WIDTH = 5)]
*** Time to filter an image of 67188864 pixels on the host = 267.829(ms)
*** CUDA kernel launch with 131072(131072) blocks of 512 threads
*** Time to filter an image of 67188864 pixels on the device = 2.548(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.458(ms)
*** Verifying the result from the device...
*** Test PASSED

[1D Gaussian filtering of image of 67188864 pixels(KERNEL WIDTH = 7)]
*** Time to filter an image of 67188864 pixels on the host = 371.966(ms)
*** CUDA kernel launch with 131072(131072) blocks of 512 threads
*** Time to filter an image of 67188864 pixels on the device = 2.614(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.437(ms)
*** Verifying the result from the device...
*** Test PASSED

[1D Gaussian filtering of image of 67188864 pixels(KERNEL WIDTH = 9)]
*** Time to filter an image of 67188864 pixels on the host = 413.614(ms)
*** CUDA kernel launch with 131072(131072) blocks of 512 threads
*** Time to filter an image of 67188864 pixels on the device = 2.548(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.407(ms)
*** Verifying the result from the device...
*** Test PASSED

[1D Gaussian filtering of image of 67188864 pixels(KERNEL WIDTH = 11)]
*** Time to filter an image of 67188864 pixels on the host = 559.378(ms)
*** CUDA kernel launch with 131072(131072) blocks of 512 threads
*** Time to filter an image of 67188864 pixels on the device = 3.405(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.478(ms)
*** Verifying the result from the device...
*** Test PASSED

*** Done

```

```

[Method 2] Started
--- GRID_REDUCTION_FACTOR = 1 ---
*** CUDA kernel launch with 65536(65536) blocks of 512 threads
*** Time to filter an image of 67188864 pixels on the device = 2.958(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.479(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 2 ---
*** CUDA kernel launch with 65536(131072) blocks of 512 threads
*** Time to filter an image of 67188864 pixels on the device = 2.584(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.466(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 4 ---
*** CUDA kernel launch with 32768(131072) blocks of 512 threads
*** Time to filter an image of 67188864 pixels on the device = 2.552(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.468(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 8 ---
*** CUDA kernel launch with 16384(131072) blocks of 512 threads
*** Time to filter an image of 67188864 pixels on the device = 2.592(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.514(ms)
*** Verifying the result from the device...
*** Test PASSED
*** Done

[Method 2] Started
--- GRID_REDUCTION_FACTOR = 1 ---
*** CUDA kernel launch with 65536(65536) blocks of 1024 threads
*** Time to filter an image of 67188864 pixels on the device = 2.732(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.708(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 2 ---
*** CUDA kernel launch with 32768(65536) blocks of 1024 threads
*** Time to filter an image of 67188864 pixels on the device = 3.123(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.492(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 4 ---
*** CUDA kernel launch with 16384(65536) blocks of 1024 threads
*** Time to filter an image of 67188864 pixels on the device = 3.324(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.409(ms)
*** Verifying the result from the device...
*** Test PASSED
--- GRID_REDUCTION_FACTOR = 8 ---
*** CUDA kernel launch with 8192(65536) blocks of 1024 threads
*** Time to filter an image of 67188864 pixels on the device = 3.235(ms)
*** Average time to filter an image of 67188864 pixels on the device = 1.365(ms)
*** Verifying the result from the device...
*** Test PASSED
*** Done

```